Remote Sensing Image Coding for Machines on Semantic Segmentation via Contrastive Learning

Junxi Zhang, Zhenzhong Chen, *Senior Member, IEEE*, and Shan Liu, *Fellow, IEEE*

***Abstract*— Due to the huge data volume of high-resolution remote sensing imagery (RSI) and limited transmission bandwidth, RSIs are typically compressed for efficient transmission and storage. However, most of the existing compression algorithms are developed based on optimizing for the human perceptual that are not suitable for remote sensing image applications where RSIs are usually used for machine interpretation tasks, such as semantic segmentation for ground–object recognition. In this article, we propose an image coding for machines (ICMs) paradigm based on contrastive learning in a fully supervised manner to boost semantic segmentation of compressed RSIs. Specifically, we build an end-to-end compression framework to make full use of the global semantic information by clustering intracategory projected embeddings and spacing intercategory embeddings apart, to compensate for the loss of feature discriminability during the compression process and reconstruct the decision boundaries between different categories. Compared to the state-of-the-art image compression methods, our proposed method significantly improves the performance of semantic segmentation on the remote sensing labeling benchmark datasets.**

***Index Terms*— Contrastive learning, image coding for machines (ICMs), remote sensing interpretation, semantic segmentation.**

I. Introduction

With the growing demand for remote sensing data, remote sensing imaging systems are rapidly developing, facilitating the application of remote sensing imageries (RSIs). On the other hand, the spatial, temporal, and radiometric resolution is continuously improving, which also leads to the exponential growth in the volume of remote sensing image data. A substantial amount of data is sent from satellites to ground stations, creating a contradiction between limited bandwidth and huge data volume. In recent years, there has been a growing interest in compressing RSIs to meet the needs of transmission and storage.

The transmitted RSIs are typically utilized for further analysis. As a fundamental problem of RSIs, semantic segmentation plays a crucial role in various applications, such as scene classification [1], road extraction [2], and change detection [3]. Manual annotation for remote sensing images is extremely time-consuming and costly. With the advancements in computer technology and the rise of deep learning, deep learning-based methods [4], [5], [6] have shown impressive performance in semantic segmentation tasks. These automatic methods can efficiently process large-scale remote sensing data. Ground stations have access to more powerful computing resources than those available on satellites, allowing for more complex and accurate image processing and interpretation. As a result, upon receiving and decoding the data on the ground stations, interpretation tasks would then be performed on the reconstructed images. However, the compressed images, especially at a high compression ratio, would get the poor performance of subsequent semantic segmentation task [7], [8], which is an urgent problem to be solved.

After decades of development, some classic compression methods commonly used in remote sensing images, such as JPEG [9], Webp [10], JPEG2000 [11], CCSDS [12], BPG [13], JPEG-XL [14], and JPEG2000 variants [15], [16], [17], [18] have achieved satisfactory coding performance. They are typically designed in a hybrid style, incorporating various hand-crafted techniques to improve coding efficiency. Different from these classic methods that rely on transformation coding, quantization coding, and other related algorithms, learned-based image compression approaches [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30] operate in an end-to-end manner. Thanks to the strong modeling capacity of neural networks, end-to-end image compression approaches have demonstrated comparable or even better performance with traditional image compression standards. This progress highlights the potential for learned-based codecs to become the next generation of high-performance and widely applicable image coding standards. Considering the extensive field of view and abundant details in remote sensing images, researchers [31], [32], [33] introduced attention mechanisms, long-range convolution, and edge map to preserve edge and texture information of compressed RSIs. However, both the traditional and the end-to-end compression approaches mentioned above are typically optimized for human perception, restricting the performance of downstream machine tasks. As shown in Fig. 1, there is a lot of misclassification in the segmentation result of the compressed RSI by the BPG codec. This issue significantly diminishes the segmentation accuracy of remote sensing images, which decreases the utilization efficiency of valuable remote sensing data.

In recent years, lots of research studies [34], [35], [36], [37] have been devoted to addressing the emerging problem of image coding for machines (ICMs) to support down-

Received 1 February 2024; revised 18 June 2024 and 7 September 2024; accepted 15 September 2024. Date of publication 14 October 2024; date of current version 30 October 2024. This work was supported in part by the National Natural Science Foundation of China under Contract 62036005 and in part by Tencent. *(Corresponding author: Zhenzhong Chen.)*

Junxi Zhang and Zhenzhong Chen are with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430072, China (e-mail: zzchen@ieee.org).

Shan Liu is with the Tencent Media Laboratory, Tencent America, Palo Alto, CA 94306 USA.

Digital Object Identifier 10.1109/TGRS.2024.3479190

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:53:04 UTC from IEEE Xplore. Restrictions apply.

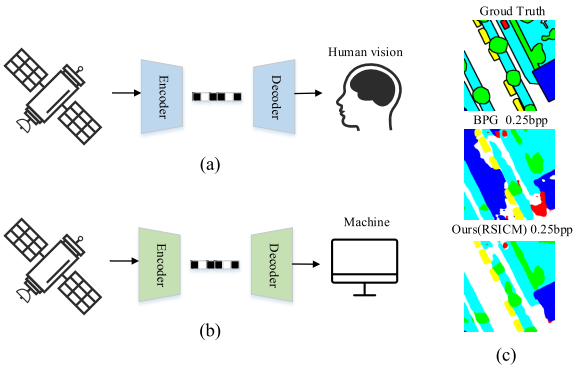


Fig. 1. (a) Image compression method for human visual. (b) Our proposed image compression for machine vision tasks. (c) Semantic segmentation results for the image from the BPG codec and RSICM (ours).

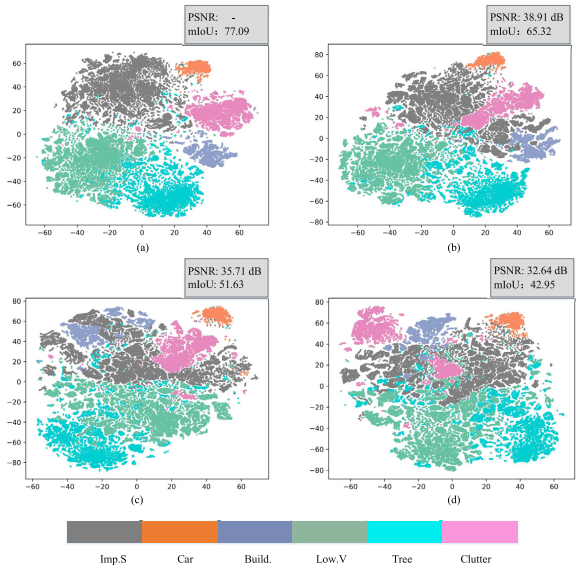


Fig. 2. t-SNE visualization of the learned features. (a) Uncompressed image. The decoded image was generated by the GMM model at three bit rates: (b) 0.45 bpp, (c) 0.17 bpp, and (d) 0.10 bpp. PSNR values and mIoU scores are marked in the upper right corner of the graphics. Features are colored according to their class.

stream machine vision tasks. Previous work [38], [39] directly extracted and transmitted high-level features that are related to the specific tasks, and subsequently, the decoded features were utilized for further analysis. However, preextracting features require substantial computing power for the sending side. Furthermore, the image signal cannot be recovered from the reconstructed features, so auxiliary manual interpretation cannot be performed on the receiving side. Other researchers [40], [41], [42], [43] improved coding efficiency by allocating more bit rates for regions of interest (ROIs). However, it needs to define and detect the ROI in advance and this operation also requires computing power for the sending side. Moreover, it often sacrifices the quality of non-ROI in exchange for the improvement of ROI, failing to achieve the global optimum. Given the occasional need for manual intervention in interpreting RSIs, these methods are not suitable for RSIs. In this article, we explore a machine vision-oriented compression method tailored explicitly for remote sensing images.

To investigate the fundamental cause of the performance degradation in the semantic segmentation task with compression, we randomly pick several images of the Potsdam val dataset and compress them by GMM [23], a compression model optimized for human perception. Then, we extract the features by ResNet-18, which is a pretrained model of PSPNet [6]. In Fig. 2, each point corresponds to the pixel embeddings after t-SNE dimensionality reduction. As we can see, the average peak signal-to-noise ratio (PSNR) score of the compressed images corresponding to Fig. 2(b) achieves 38.91 dB, which means almost imperceptible differences between the compressed images and the uncompressed image for human vision. However, the mean Intersection over Union (mIoU) value decreases from 77.09 to 65.32, thus directly employing these compressed images for semantic segmentation tasks is not appropriate. As Fig. 2(a) shows, the feature space corresponding to the uncompressed images exhibits obvious intraclass compactness and interclass dispersion. This characteristic is considered to be indicative of a good embedding space for semantic segmentation [44]. However, coincident with the increase in compression ratio is the increase of ambiguity in feature space and a gradual decline in segmentation accuracy. From our perspective, the main reason is that the compression network regards mean squared error (mse) loss as the only measure of distortion. It prioritizes signal fidelity in the pixel domain while neglecting potential loss in the feature domain. Thus, when the bit rate is relatively high, even though the changes in the pixel domain may be minor to human eyes, some changes have taken place within the feature domain, damaging the above-mentioned characteristic beneficial for segmentation would be damaged, resulting in decreased segmentation accuracy. To address this issue, we propose a remote sensing image coding framework for machines from a unique perspective of enhancing feature discrimination.

To facilitate the machine vision applications of compressed RSIs, in this article, we propose RSICM, a novel contrastive learning-based remote sensing ICM framework, which enables efficient image compression for the semantic segmentation task. Based on the observation of the ambiguity in the feature domain of compressed images, this framework utilizes contrastive learning to effectively constrain the feature space, enhancing feature discrimination. Specifically, compressed images are transferred from the pixel domain to projected embedding space, with same-category embeddings as positive samples and different-category embeddings as negative samples. Then, we introduce contrastive learning to guide the compression network to preserve the feature characteristics that benefit semantic segmentation, namely intraclass compactness and interclass dispersion.

The main contributions of our work are listed as follows.

1) We explore the impact of remote sensing image compression on semantic segmentation and observe that compression affects the features discriminability of the compressed images. To address this issue, we propose a remote sensing ICM framework from a unique perspective of feature discrimination.
2) We integrate contrastive learning into the rate-distortion (RD) optimization process, guiding the compression

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:53:04 UTC from IEEE Xplore. Restrictions apply.

network to enhance intraclass compactness and interclass dispersion, thereby compensating for the feature discrimination distorted by the encoding process.

3) Our method provides a solution that can be applied to various learning-based image compression algorithms. Implementations of two common algorithms validate their flexibility and efficiency. Extensive experimental results demonstrate that the proposed method reconstructs the decision boundaries between different classes and achieves superior efficient image compression for semantic segmentation.

The remainder of this article is organized as follows. Related work is briefly introduced in Section II. Section III presents the framework of RSICM. Detailed experimental configurations and evaluations on the ISPRS 2-D labeling benchmark and LoveDA datasets are presented in Section IV. Finally, conclusions are drawn in Section V.

II. Related Work

A. Deep Learning-Based Image Coding

In recent years, learned image compression has achieved many breakthroughs. Pioneering work by Ballé et al. [19] introduced an autoencoder structure based on generalized divisive normalization (GDN), transforming input images into latent representations. To address the nondifferentiability issue in quantization, they incorporated uniform distribution noise into the training process to simulate quantization. Nevertheless, spatial dependencies persisted within the latent space. On this basis, Ballé et al. [20] introduced the HyperPrior to capture spatial redundancy, which is incorporated into the bitstream as side information, allowing for a conditional entropy model. After that, a mean and scale Gaussian mixture model and autoregressive context model [21] were introduced to improve the coding efficiency. To further improve the RD performance, prior work introduces attention modules [23], [26], invertible structures [27], or transformer architectures [28], [29] to the main analysis, synthesis networks, and entropy model. ELIC [30] proposed uneven channel-conditional adaptive coding to balance the compression performance and running speed. Considering the extensive field of view and abundant details in remote sensing images, some methods made improvements based on the end-to-end compression framework. Zhao et al. [31] proposed a GAN-based compression model and introduced an enhanced Laplacian of Gaussian loss to enhance the edges, contours, and textures of remote sensing images. Xiang and Liang [33] proposed a long-range convolution to capture context information along with the nonlocal attention model to reduce computation complexity. However, all the aforementioned works are optimized for human perception, resulting in a significant degradation in performance when the reconstructed images are deployed for machine tasks.

B. Image Coding for Machines

Nowadays, machine-to-machine (M2M) communication is merging, where compressed images are consumed by machines for vision tasks [45]. To facilitate M2M scenarios, the Moving Picture Experts Group (MPEG) introduced an ad hoc group on video coding for machines (VCMs) in 2019. This standard includes two tracks, with one paradigm being *analyze-then-compress*. Features are extracted, then coded into bin streams on the sending end, and transmitted. The reconstructed features are directly employed for intelligent tasks on the receiving side. The previous work [38], [39], [46] presented lossy/lossless intermediate deep features compression using the high-efficiency video coding (HEVC) standard. In contrast to utilizing a pretrained network for feature extraction, Singh et al. [47] jointly optimized the task network along with rate loss to achieve consistent improvement. In another paradigm referred to *compress-then-analyze*, the images are compressed and transmitted to the receiving side, where both the feature extraction and analysis are executed. Previous work [40], [41], [42], [43], [48] proposed bit allocation strategies based on ROI, which is manually specified or generated from the object detector. Fischer et al. [49] incorporated feature-based distortion metrics into the RDO process of VTM-8.0. However, the intricate interdependence among various modules in traditional codecs presents challenges when aiming to optimize the entire system jointly. To achieve end-to-end optimization, some work [35], [36], [50] has explored the feasibility of end-to-end image compression for machine vision, providing a reference for subsequent research. Furthermore, Fischer et al. [51] introduced some layers of a pretrained analysis network to create masks for concealing elements within the latent space during compression. The first track imposes a substantial computational burden on the sending side due to the necessity of preextracting image features, which renders the application of this paradigm to onboard compression of remote sensing images impractical. Furthermore, the challenge of recovering the image signal from its features persists. The compressed image signal from the latter paradigm remains interpretable for humans, allowing for manual intervention when needed and promoting transparency in decision-making. In conclusion, the latter paradigm is more suitable for remote sensing coding for machines. In this article, we focus on the latter paradigm and introduce contrastive learning to compensate for the feature discriminability distorted by the encoding process.

C. Remote Sensing Semantic Segmentation

Semantic segmentation stands as a fundamental task within the realm of remote sensing image interpretation. Long et al. [4] replaced the fully connected layer of the network with the convolutional layer, resulting in the pioneering end-to-end pixel-wise prediction framework known as the fully convolutional network (FCN). However, the FCN struggled to efficiently integrate feature information from both high and low layers of the image, leading to low semantic segmentation accuracy. In response to this challenge, an encoder–decoder structure was introduced. U-Net [52] introduced skip connections, establishing a linkage between the feature maps from the encoder and their counterparts from the decoder. This strategy combined deep semantic insights with shallow spatial positional information. SegNet [5] utilized maximized

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:53:04 UTC from IEEE Xplore. Restrictions apply.

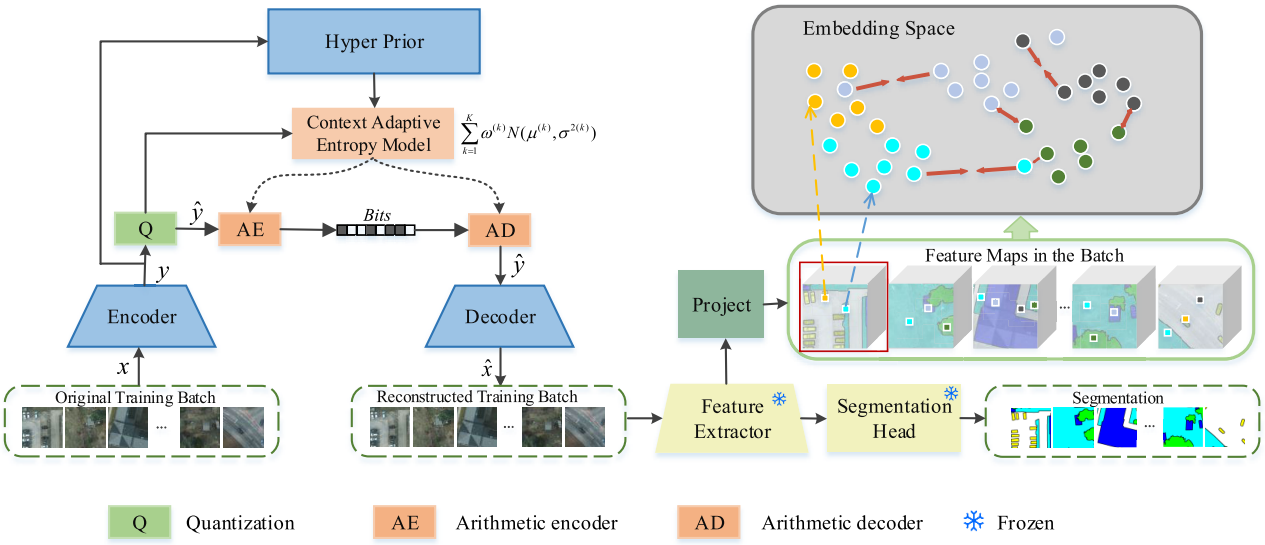


Fig. 3. Illustration of our proposed RSICM. The color point indicates an embedding anchor of each pixel. Different colors denote different categories. In the embedding space, projected embeddings belonging to the same category are pulled closer, while those from different categories are pushed apart.

pooling indices to guide the upsampling process during decoding. This approach enhanced segmentation while effectively reducing the number of model parameters. To enlarge the limited receptive field, Yu and Koltun [53] proposed dilated convolution to capture contextual information across extensive regions without introducing additional parameters. PSPNet [6] employed a pyramid pooling module to aggregate multiscale contextual information effectively. The DeepLab series [54], [55], [56] introduced the atrous spatial pyramid pooling (ASPP) module, employing diverse atrous rates to capture multiscale contextual information effectively. This approach aimed to address the challenge posed by targets of various scales. Mi and Chen [57] introduced superpixel segmentation with region loss, which can enhance the homogeneity with and heterogeneity between superpixels to strengthen the edges of ground objects. Considering the large-scale presence of small targets and indistinct boundaries inherent to remote sensing images, some work [58], [59], [60] introduced an attention mechanism. Furthermore, several studies [61], [62], [63] have proposed approaches rooted in the transformer architecture. However, when the input is compressed images, the accuracy of semantic segmentation will be greatly reduced.

III. Method

In our exploratory experiments, we observe the ambiguity within the feature domain of compressed images. To mitigate this issue, we creatively incorporate contrastive learning into an end-to-end compression framework. The framework of our work is as depicted in Fig. 3, images are encoded into a binary stream and stored for transmission. Upon receiving, a decoder reconstructs these images for segmentation. The reconstructed images are initially processed by the feature extractor to obtain deep features. Then, a project layer projects these features into an embedding space, where contrastive learning is applied. Through contrastive learning strategy, the compression framework can acquire the capacity to retain the intraclass compactness and the interclass dispersion within feature space, which is vital for semantic segmentation. To verify the effectiveness of our proposed RSICM, we implement it on two specific networks, GMM [23] for learned image compression and PSPNet [6] for semantic segmentation. It is worth noting that our approach is not constrained to particular pairings of compression and segmentation networks. It can be implemented on arbitrary combinations.

A. Problem Formulation

In this section, we mathematically formulate the problem addressed in this article. We define x as the input RSIs. In contrast to hand-crafted transformation modules in hybrid image codecs, the input image is mapped to a latent representation via parametric analysis transforms in learning-based image compression. Then, the representation undergoes quantization, yielding a discrete-valued vector, which is lossless-compressed into a bitstream using various coding techniques like Huffman or Arithmetic coding. On the receiving side, the reconstructed image is derived from the discrete-valued through inverse synthesis transforms

$$\hat{x} = g_s(Q(g_a(x;\phi));\theta) \tag{1}$$

where ϕ denotes the parameters of the analysis transforms $g_a(\cdot)$ and θ denotes the parameters of the inverse synthesis transforms $g_s(\cdot)$. Q represents the quantization process. During the training, the quantization is approximated by uniform noise $\mathcal{U}(-(1/2),(1/2))$, which enables gradients to propagate through backpropagation. During the reference, the quantization is conducted by the round function.

To eliminate spatial dependencies within the latent representation y, the HyperPrior was first introduced by [20]. It introduced side information z, which is shared between the encoder and the decoder. Cheng et al. [23] proposed the Gaussian mixture model to enable the modeling of arbitrary likelihoods

$$\hat{z} = h_s(Q(h_a(x;\phi_h));\theta_h) \tag{2}$$

$$p_{\hat{y}|\hat{z}}(\hat{y}\mid\hat{z}) \sim \sum_{k=1}^{K}\omega^{(k)}N(\mu^{(k)},\sigma^{2(k)}) \tag{3}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:53:04 UTC from IEEE Xplore. Restrictions apply.

where ϕ_h and θ_h denote the parameters of analysis transforms $h_a(\cdot)$ and inverse synthesis transforms $h_s(\cdot)$ in HyperPrior. $\hat{y}$ is the quantized latents. $\hat{z}$ is the quantized Hyperlatents. k denotes the index of mixtures of the Gaussian mixture model.

After compression, the reconstructed images $\hat{x}$ are fed into the semantic segmentation network to obtain segmentation results. As illustrated in Fig. 3, the reconstructed images are first fed into *feature extractor* to generate dense features, which are subsequently transformed into segmentation results by *segmentation head*

$$S = f_{\text{SEG}}(f_{\text{CNN}}(\hat{x}; \beta), \varphi) \tag{4}$$

where β and φ denote the parameters of the feature extractor f_{CNN} and f_{SEG}, respectively.

If the parameters β and φ can be jointly optimized alongside the compression framework, there will exist various vision models for distinct levels of image quality. In the scope of this article, we explore a coding framework for machines. This implies that RSIs compressed at different bit rates using the framework should be directly input into a pretrained semantic segmentation model, eliminating the need to retrain vision models for varying levels of image quality. To realize this objective, we assume that the downstream visual task network remains unaltered as compression ratios change. So, the optimization function of our framework can be described as

$$\arg\min_{\phi,\theta,\phi_h,\theta_h} \lambda \cdot \mathcal{L} + \mathbb{E}_{x\sim p_x}[-\log_2 p_{\hat{y}}(\hat{y})] + \mathbb{E}_{x\sim p_x}[-\log_2 p_{\hat{z}}(\hat{z})] \tag{5}$$

where λ represents the Lagrange multiplier, and learned-based compression models use a particular set of λ values to achieve the desired tradeoff between bitrate and distortion. $\mathcal{L}$ comprises both pixel distortion and contrastive loss, which will be detailed in Section III-B. The second and third terms represent the rate of latents and quantized Hyperlatents, respectively.

B. Image Coding Framework via Contrastive Learning

As analyzed in Section I, uncompressed images exhibit clear intraclass compactness and interclass dispersion, considered crucial for a robust segmentation embedding space [44]. With an increase in compression ratio, embeddings belonging to different classes become entangled, resulting in segmentation performance degradation of compressed images. To make up for the loss of feature discrimination, we incorporate contrastive learning to supervise the compression network.

In unsupervised learning, images are augmented through cropping, flipping, rotation, coloring, or other transformations to generate positive samples, with other images in the same mini-batch considered as negative samples. However, this sample selection strategy is unsuitable for addressing the specific problem discussed in Section I. This is because there is a potential scenario where embeddings belonging to the same category could be considered as negative samples. In such a scenario, the clustering of intraclass features would be adversely affected, deviating from the initial concept of compensating for the loss of intraclass compactness and interclass dispersion during the compression process.

To address this issue, inspired by the mechanism of supervised contrastive learning [44], [64], we propose a contrastive learning-based ICMs framework. Specifically, we identify pixel embeddings of pixels belonging to the same category as positive samples and those from the different categories as negative samples. Furthermore, to broaden the scope of comparisons, we search for positive and negative samples in a mini-batch, jumping out of the limit of one picture. With the label information, we build the category relevance. Cosine similarity is used to measure the similarity between pixel embeddings. Therefore, the contrastive learning loss function in this article can be defined as

$$\mathcal{L}_i^{\text{NCE}} = \frac{1}{|\mathbb{P}|} \sum_{e_j^+ \in \mathbb{P}} -\log \frac{P_{(e_i,e_j^+)}}{P_{(e_i,e_j^+)} + N_{(e_i,e_j^-)}} \tag{6}$$

$$P_{(e_i,e_j^+)} = \exp(e_i \cdot e_j^+/\tau) \tag{7}$$

$$N_{(e_i,e_j^-)} = \sum_{e_j^- \in \mathbb{N}} \exp(e_i \cdot e_j^-/\tau) \tag{8}$$

where e_i denotes the pixel embedding vector corresponding to the semantic label c. The collection $\mathbb{P}$ consists of embeddings sharing the same label, whereas $\mathbb{N}$ comprises embeddings with different labels. The hyperparameter τ represents a scale temperature factor regulating the smoothness or sharpness of the probability distribution produced by the softmax function.

The optimization objective of the loss function is to enhance intraclass compactness and interclass dispersion. By incorporating this into the RD optimization process, the compression network is guided to preserve the feature distribution, thereby compensating for the feature discrimination loss caused by compression.

C. Training Strategy

Image compression is often modeled as an RDO problem, as formulated in (5). Existing compression methods typically use mse loss as the only measure of distortion. The mse loss only quantifies the pixel-wise error between the uncompressed and reconstructed images, thereby failing the compression network to capture higher-level semantic information. To effectively leverage the semantic information while accounting for the bit rate, we extend the concept of distortion to the feature domain in RDO optimization. $\mathcal{L}$ defined in (5) is formulated as

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} (x_i - \hat{x}_i)^2 + \mathcal{L}_i^{\text{NCE}}. \tag{9}$$

Algorithm 1 outlines the training strategy details. The images $x \in \mathbb{R}^{N\times 3\times H\times W}$ undergo compression by the end-to-end compression network $f(\cdot)$, resulting in $\hat{x} \in \mathbb{R}^{N\times 3\times H\times W}$ with N, H, and W denoting batch size, height, and width of the images, respectively. These compressed images are initially processed by the *feature extractor*, denoted as $g_b(\cdot)$ to extract deep features. During extraction, the input images are transformed and downsampled into corresponding features $I \in \mathbb{R}^{N\times 512\times H/8\times W/8}$. Subsequently, the *project* layer,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:53:04 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 RSICM: PyTorch Pseudocode

Input: A batch of images, Lagrange multiplier λ
Output: Parameters of end-to-end image compression network

```
# g: end-to-end compression network
# f_CNN: ResNet-18 backbone  h: project head
# ℙ, ℕ: Collections of positive and negative samples
# N: batch size  W, H: width and height of images
x̂ = g(x)
I = f_CNN(x̂)
e = h(I)
for all i ∈ {1, ..., N × W × H} do
  for all j ∈ {1, ..., N × W × H} do
    if C(e_i) == C(e_j) then
      ℙ.append(e_j)
    else
      ℕ.append(e_j)
    end if
  end for
  ℙ = rand_sample_K(ℙ)
  ℕ = rand_sample_K(ℕ)
  f_CNN.detach()
  define
  L_i^NCE = (1/K) Σ_{e_j^+ ∈ ℙ} − log exp(e_i·e_j^+/τ) / (exp(e_i·e_j^+/τ) + Σ_{e_j^- ∈ ℕ} exp(e_i·e_j^-/τ))
  L = R + λ(L_mse + L^NCE(e_i, e_j^+, e_j^-))
  L.backward()
  update networks g and h to minimize L
end for
return g(·), remove h(·)
```

denoted as $h(\cdot)$, maps these features into an embedding space, where contrastive learning is applied. The projection layer consists of convolution layers with 1×1 kernel size, batch normalization, and ReLU activation function as shown in Fig. 4. Following projection, the dense features are further mapped into embeddings $z \in \mathbb{R}^{N\times 256\times H/8\times W/8}$. For a given pixel embedding z_i, represented as a 256-D vector, we select its positive and negative samples within the mini-batch based on their labels, forming collections $\mathbb{P}$ and $\mathbb{N}$. We then randomly extract K samples from each of these two collections. During training, we optimize the end-to-end compressed networks $f(\cdot)$ and *project* head $h(\cdot)$ to minimize the loss function, while the pretrained semantic segmentation network PSPNet, downloaded from *OpenMMLab*,[1] is frozen to prevent its participation in backpropagation. Following training, we remove the *project head*, preserving only the compression network for remote sensing image coding. This strategy ensures the feasibility of the assumption stated in Section III-A.

IV. Experimental Results and Analysis

In this section, we validate the effectiveness of the proposed RSICM. The details of the experiments are described in Sections IV-A–IV-D. Experimental results and analysis are shown in Section IV-E, which validate that our method can improve the coding efficiency for the semantic segmentation task.

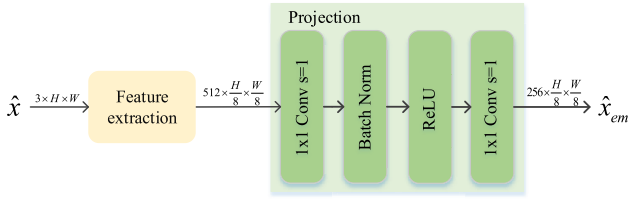


Fig. 4. Illustration of the projection architecture.

A. Experimental Settings

Following the setting of CompressAI,[2] we select eight sets of λ values (λ = 117, 228, 436, 845, 1625, 3141, 6060, and 11 705) to regulate the RD curve. To align the four bit rates displayed in Table I, we add eight λ values: (λ = 371, 1496, 5006, and 17 557) for the Potsdam dataset and (λ = 143, 650, 2471, and 10 404) for the Vaihingen dataset. Due to the limited volume of the dataset, we loaded pretrained models from CompressAI and fine-tuned them using the Potsdam or Vaihingen dataset to obtain more stable results. The network optimized by mse loss serves as the reference learned image compression approach in the remainder of this article. To ensure a fair comparison, each model was retrained on the datasets. The experimental results demonstrate that incorporating fine-tuning has yielded more efficient coding. The batch size is set to 16. We employed the Adam optimizer with β_1 set to 0.9 and β_2 set to 0.999. The initial learning rate is set to 1×10^{-4} and halved when the evaluation loss plateaus for 20 epochs.

B. Evaluation Metrics

1) mIoU and mFscore: To assess the quantitative performance of the semantic segmentation task, we employ two metrics, that is, the mIoU and mean F-score (mFscore)

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{10}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{11}$$

$$\text{mIoU} = \frac{1}{M}\sum_{m=1}^{M}\frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \tag{12}$$

$$\text{mFscore} = \frac{1}{M}\sum_{m=1}^{M}\frac{2\cdot\text{Precision}\cdot\text{Recall}}{\text{Precision}+\text{Recall}} \tag{13}$$

where TP, FP, and FN are the number of true positives, false positives, and false negatives, respectively. M is the number of classes.

2) BD-rate: There is always a tradeoff between bits saving and the quality of the reconstructed image. Consequently, Bjøntegaard Delta Rate (BD-rate) [65] is introduced to compare the efficiency of two codecs comprehensively. BD-rate quantifies the bits saving while maintaining the same image

[1] https://github.com/open-mmlab

[2] https://github.com/InterDigitalInc/CompressAI

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:53:04 UTC from IEEE Xplore. Restrictions apply.

TABLE I

MIoU AND MFSCORE VALUES OF THE PROPOSED COMPRESSION METHOD AND OTHER COMPRESSION METHODS ON THE ISPSR BENCHMARK. MIoU/MFscore VALUES OF UNCOMPRESSED IMAGES ARE 77.09/85.62 FOR POTSDAM AND 71.46/81.75 ON THE VAIHINGEN DATASET. BOLD FONTS INDICATE THE BEST PERFORMANCE

Method	Potsdam								Vaihingen							
	mIoU				mFscore				mIoU				mFscore			
	0.1	0.25	0.5	1.0	0.1	0.25	0.5	1.0	0.1	0.25	0.5	1.0	0.1	0.25	0.5	1.0
JPEG [9]	-	19.36	65.91	75.71	-	30.44	78.05	84.41	-	41.38	61.98	69.65	-	56.6	74.71	80.5
WebP [10]	34.98	52.61	67.35	75.55	48.1	66.43	78.71	84.58	-	60.86	67.52	70.25	-	74.48	79.37	80.95
JPEG2000 [11]	35.98	52.48	68.08	75.45	50.98	66.95	79.41	84.55	40.04	57.71	66.8	69.95	55.24	71.85	78.69	80.75
BPG [13]	35.39	56.19	67.66	74.42	47.61	69.33	78.95	83.86	53.92	62.82	68.24	70.63	69.05	76.1	79.77	81.26
JPEG-XL [14]	36.03	61.6	70.67	75.7	46.63	74.21	81.31	84.67	-	57.25	69.68	**71.53**	-	71.67	80.78	**81.94**
GDN [19]	35.67	51.47	67.21	76.12	47.78	65.14	78.66	85.12	56.48	62.46	67.23	70.11	70.66	75.69	79.09	80.91
Hyper [20]	39.58	56.51	70.21	76.4	52.22	69.76	80.99	85.27	56.79	63.34	67.84	70.31	71.13	76.44	79.47	81.03
Joint [21]	40.66	57.62	70.64	76.6	53.38	70.67	81.21	85.37	58.45	64.59	68.11	70.66	72.5	77.35	79.66	81.26
GMM [23]	42.95	60.34	72.06	76.51	56.53	71.29	82.23	85.29	60.17	64.34	68.9	70.53	74.02	77.13	80.08	81.15
Invcomp [27]	40.58	60.45	70.44	76.43	53.55	73.13	81.03	85.26	60.28	64.68	69.03	70.5	74.04	77.31	80.18	81.12
TBTC [29]	47.51	61.41	72.57	76.76	61.3	73.81	82.65	85.47	61.33	64.98	69.51	70.93	74.81	77.51	80.46	81.39
STF [26]	47.78	60.82	71.59	76.40	61.6	73.33	81.88	85.25	61.02	65.44	69.08	70.61	74.6	77.85	80.17	81.20
ELIC [30]	47.28	62.45	74.42	76.56	61.19	74.74	82.54	85.32	61.05	65.46	69.14	70.8	74.56	77.89	80.25	81.35
TCM [28]	47.75	62.09	72.53	76.66	61.53	74.36	82.61	85.4	60.52	64.65	68.91	70.73	74.16	77.25	80.07	81.29
RSICM (Joint)	56.11	69.84	74.02	**76.77**	68.31	80.37	83.63	**85.50**	61.58	66.41	69.29	70.92	74.61	78.29	80.37	81.41
RSICM (GMM)	**61.42**	**72.87**	**75.77**	76.76	**73.33**	**82.67**	**84.78**	85.44	**64.12**	**68.22**	**69.68**	71.09	**76.5**	**79.57**	**80.78**	81.52

quality with an anchor. Therefore, a negative BD-rate value indicates that the codec has better compression efficiency than the anchor. The smaller the BD-rate the higher the compression efficiency of the corresponding method. Typically, PSNR as an evaluation metric is obtained to measure the performance of different codecs for the human visual system. In the application scenario of this article, PSNR is not the primary metric. In this article, we incorporate mIoU and mFscore as the primary metrics.

C. Data Description and Preprocessing

The proposed compression framework is evaluated on the ISPRS 2-D semantic labeling benchmark and LoveDA dataset [66]. The ISPRS benchmark includes two subsets: the Potsdam dataset and the Vaihingen dataset, comprising ground-truth labels for six classes: *impervious surfaces, buildings, low vegetation, trees, cars, and clutter*. The Potsdam dataset is a high-resolution dataset consisting of 38 orthorectified tiles with a resolution of 6000×6000 pixels. It comprises 4-band IRRGB aerial images. Notably, the proposed method only utilizes the RGB bands. The Vaihingen dataset consists of 33 orthorectified images. It is a 3-band dataset incorporating IRRG (infrared, red, and green) bands. For the ISPRS benchmark, according to the official data split, the Potsdam dataset consists of 24 training images and 14 testing images, while the Vaihingen dataset includes 16 training images and 17 testing images. We randomly select two images from the training images for validation and the remaining images for training. The original image tiles were cropped into 512×512 pixel patches in the experiments. The LoveDA dataset consists of 5987 optical remote sensing images with a spatial resolution of 0.3 m, containing red, green, and blue bands. Specifically, 2522 images are used for training, 1669 images for validation, and the officially provided 1796 images for testing. The dataset is collected from two scenes: urban and rural, covering seven land-cover categories. It is considered challenging due to three key characteristics: multiscale objects, complex background, and inconsistent distribution.

TABLE II

MIoU AND BD-RATE (%) WITH JPEG2000 ANCHOR ON LOVEDA. MIoU VALUES OF UNCOMPRESSED IMAGES IS 47.17

Method	mIoU				BD-rate	
	0.1	0.25	0.5	1.0	mIoU	PSNR
JPEG2000 [11]	19.00	25.61	31.99	38.61	0	0
WebP [10]	18.99	27.16	32.65	37.68	-5.99	13.40
BPG [13]	23.21	31.14	35.76	39.71	-40.25	-28.22
JPEG-XL [14]	21.56	31.74	36.73	41.97	-46.07	-2.11
GMM [23]	28.89	40.38	45.14	47.29	-77.61	-50.06
TBTC [29]	31.04	41.09	46.51	48.19	-80.06	-52.57
STF [26]	31.40	40.81	45.47	47.64	-79.61	-52.72
ELIC [30]	30.39	42.13	46.29	47.73	-80.77	-52.81
TCM [28]	32.92	41.47	45.93	47.67	-82.17	**-54.19**
RSICM(GMM) [9]	**37.94**	**45.14**	**47.63**	**48.26**	**-89.92**	-49.20

D. Comparison Methods

To evaluate the performance of our method, we choose 14 models, including both the traditional hybrid coding framework and the learning-based coding framework. The highlights of these methods can be outlined as follows.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:53:04 UTC from IEEE Xplore. Restrictions apply.

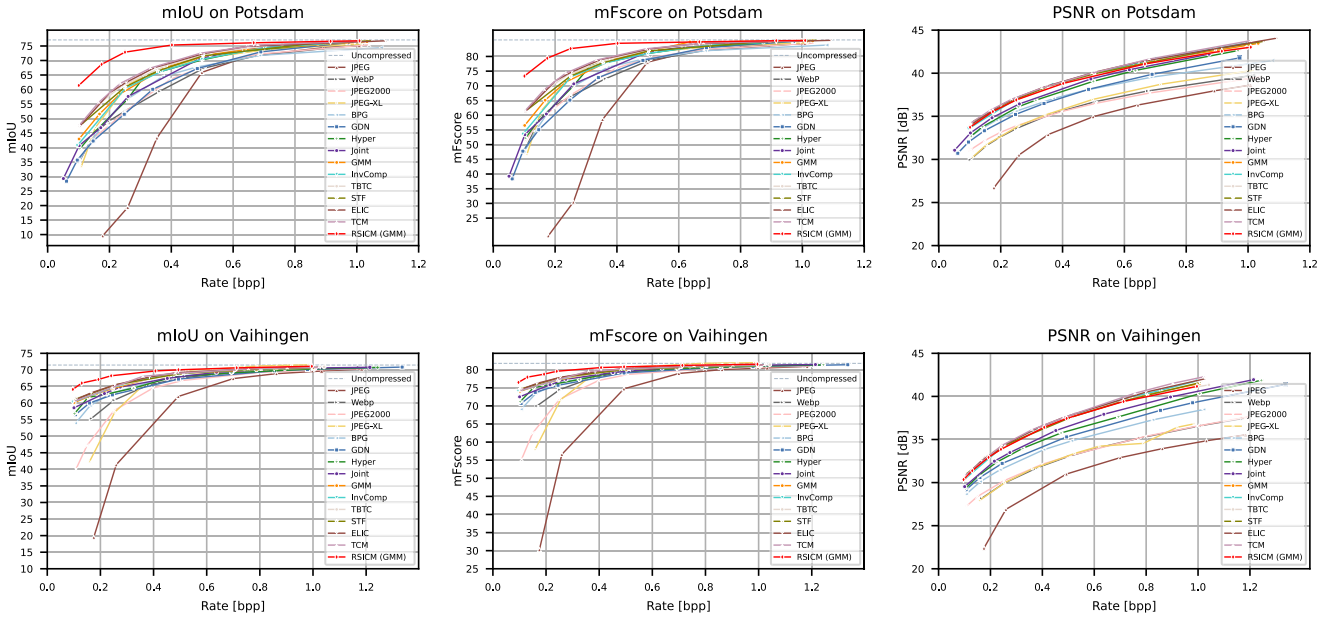


Fig. 5. RD curve on the ISPRS 2-D labeling dataset using mIoU, mFscore, and PSNR as the quality metrics.

1) Traditional Coding Framework:

1) *JPEG [9]:* A widely used compression method.
2) *WebP [10]:* A raster graphics file format.
3) *JPEG2000 [11]:* A standard based on wavelet transform.
4) *BPG [13]:* Based on intraframe technology of HEVC.
5) *JPEG-XL [14]:* The next-generation image compression codec by the JPEG committee.

2) Learning-Based Coding Framework:

1) *GDN [19]:* GDN layers for nonlinear transforms.
2) *Hyper [20]:* HyperPrior as additional information.
3) *Joint [21]:* Autoregressive context entropy model.
4) *GMM [23]:* Gaussian mixture model and attention model.
5) *InvComp [27]:* Invertible encoding network.
6) *TBTC [29]:* Based on transformer architectures.
7) *STF [26]:* Window-based attention model.
8) *ELIC [30]:* Uneven channel-conditional adaptive coding.
9) *TCM [28]:*: Transformer-based entropy model.

It is worth noting that we fine-tune all the end-to-end comparison models with mse loss with the same configuration described in Section IV-A to ensure fairness.

E. Experimental Results and Analysis

1) Comparison on mIoU and mFscore Values: To quantitatively assess the effectiveness of our method, mIoU and mFscore values for various comparison methods on the ISPRS benchmark and the LoveDA dataset across different bit rates are tabulated in Tables I and II, respectively. It is worth noting that the bit rate fluctuates within plus or minus 5% of the given bit rate point. JPEG fails to reach 0.1 bpp on all datasets. WebP and JPEG-XL fail on the Vaihingen dataset. It is evident that our method greatly improves the accuracy of semantic segmentation tasks at a similar bit rate. Compared to GMM, the basic model of our method, our method only needs half the bit rate to achieve basically the same performance on segmentation. Compared to more efficient methods, that is, ELIC, TCM, and so on, our method achieves higher coding efficiency on both the ISPRS benchmark and the LoveDA dataset when evaluating segmentation performance. However, as the bit rate approaches the relatively high bit rate, the performance gap between our method and other algorithms is becoming small. The performance of JPEG-XL on the Vaihingen dataset under 1 bpp surpasses our method. This can be attributed to a saturation of performance at high bitrates [35], [36]. To verify the effectiveness of our method across different compression algorithms, we directly conduct it on another baseline network, Joint [48]. The results on the ISPRS benchmark presented a substantial improvement in segmentation performance achieved by our method. This provides evidence that our method can be easily implemented on other learning-based compression algorithms, resulting in an improved balance between coding bitrate and machine analysis performance.

2) Comparison on the RD Curve: To further verify the effectiveness of the proposed method, supplementary experiments were conducted. Fig. 5 illustrates the results of mIoU, mFscore, and PSNR values on the ISPRS benchmark across various bit rates, spanning from 0.1 to 1.2 bpp. The rate is measured by bits per pixel (bpp). Our approach demonstrates superior RD performance over alternative techniques across all bit rates as measured by mIoU and mFscore. The TCM model, refined with mse loss on the training dataset, exhibits the best RD performance when evaluated using the PSNR. It is important to note that our method exhibits substantial improvements in RD performance as measured by mIoU and mFscore.

3) Comparison on BD-rate: The corresponding BD-rate result on the ISPRS benchmark with JPEG as the anchor is listed in Table III. As can be seen, our method achieves 66.87%/65.87% bits saving over JPEG on average for the same

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:53:04 UTC from IEEE Xplore. Restrictions apply.

TABLE III

BD-RATE (%) RESULTS FOR THE CURVES SHOWN IN FIG. 5 WITH THE ANCHOR JPEG. BOLD FONTS INDICATE THE BEST PERFORMANCE

Method	Potsdam			Vaihingen		
	mIoU	mFscore	PSNR	mIoU	mFscore	PSNR
WebP [10]	-12.52	-12.40	-29.96	-37.67	-41.53	-32.54
JPEG2000 [11]	-17.26	-17.80	-27.89	-29.87	-31.97	-33.85
BPG [13]	-19.06	-19.04	-51.34	-41.42	-44.12	-52.87
JPEG-XL [14]	-27.80	-27.56	-33.15	-42.60	-44.53	-33.70
GDN [19]	-19.19	-18.67	-49.27	-35.81	-39.29	-57.81
Hyper [20]	-30.03	-30.23	-56.36	-42.38	-45.23	-62.96
Joint [21]	-29.25	-28.55	-60.06	-40.55	-43.27	-65.58
GMM [23]	-30.12	-30.00	-64.63	-50.23	-51.64	-71.62
InvComp [27]	-27.12	-26.65	-65.66	-50.23	-52.15	-72.36
TBTC [29]	-33.17	-32.46	-65.63	-53.81	-54.71	-73.40
SFT [26]	-30.43	-29.67	-65.48	-51.13	-51.81	-72.53
ELIC [30]	-34.59	-34.08	-66.23	-53.00	-54.26	-72.92
TCM [28]	-37.13	-36.31	**-66.70**	-51.46	-52.31	**-73.46**
RSICM (Joint)	-53.69	-52.91	-59.60	-52.08	-53.48	-65.29
RSICM (GMM)	**-66.87**	**-65.87**	-64.27	**-67.76**	**-65.14**	-71.51

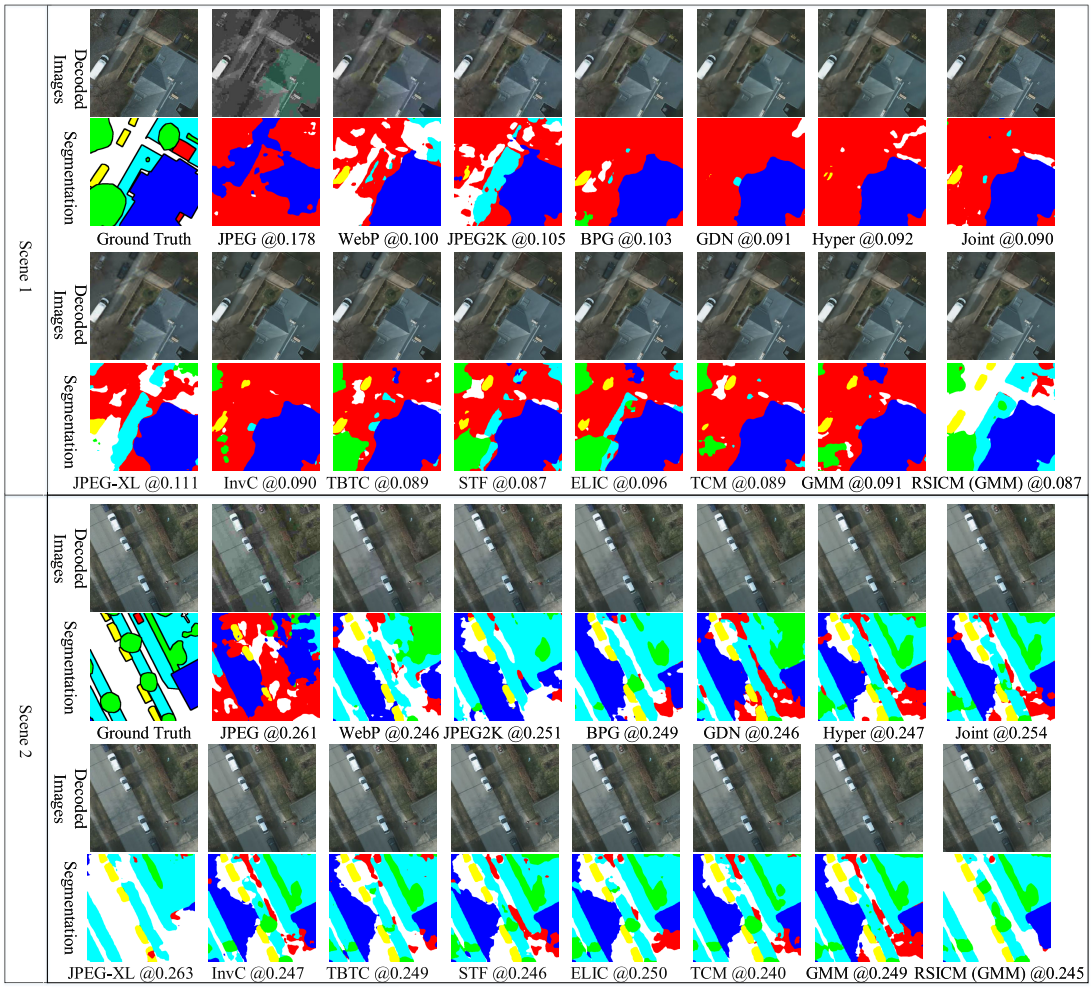


Fig. 6. Visual results of several codecs on the Potsdam dataset, including decoded images and segmentation results. The bpp of each compressed image is labeled below the image. Our method yields more accurate results with the sharpest edges.

mIoU/mFscore value on the Potsdam dataset. Meanwhile, our method achieves 67.76%/65.14% bits saving on the Vaihingen dataset. As the primary metric of the application scenario in this article, it is evident that our method is superior to all comparisons. When concerning PSNR, our method results in 64.27%/71.51% bits saving, in comparison

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:53:04 UTC from IEEE Xplore. Restrictions apply.

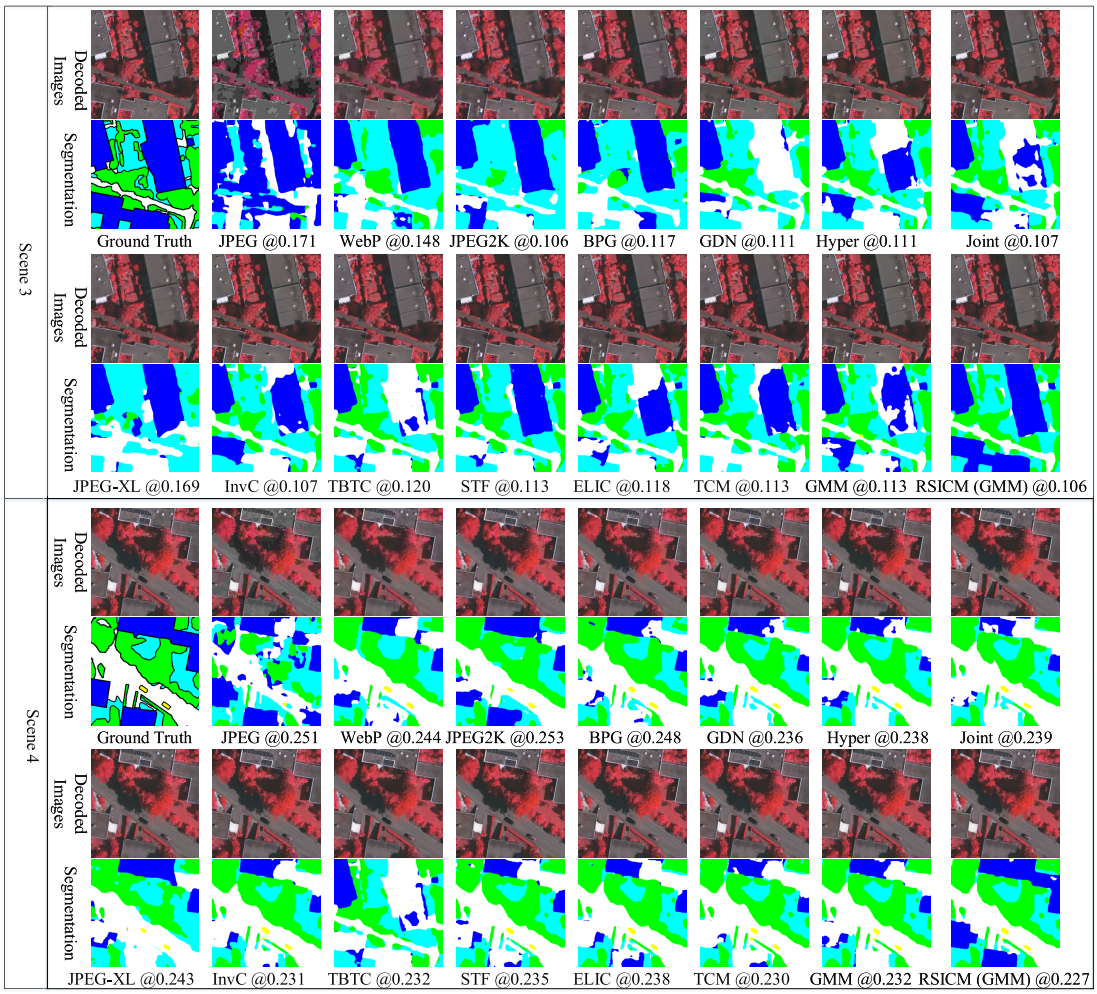


Fig. 7. Visual results of several codecs on the Vaihingen dataset.

with the GMM model with 64.63%/71.62% bits saving. The LoveDA dataset is challenging due to the multiscale objects, complex background, and inconsistent distribution [66]. The segmentation results for compressed images using JPEG are significantly low. Consequently, the BD-rate result on LoveDA with JPEG2000 as the anchor is listed in Table II. The BD-rate result indicates that our method can achieve 82.92% bitrate saving on average under equivalent segmentation performance.

4) Visualization of Semantic Segmentation Results: For a more intuitive comparison of the segmentation results, we visualized four scenes of the ISPRS benchmark at 0.1 and 0.25 bpp, respectively. Due to the lack of annotation on test images, visualization for the LoveDA dataset is unavailable. The segmentation results and the reconstructed images are presented in Figs. 6 and 7. When the segmentation network interprets the reconstructed images compressed by algorithms optimized for human perception, the boundaries between different categories become blurry, and misclassification is widespread. As shown in Fig. 2, as the bit rate decreases, impervious surfaces and clutter are initially mixed in the feature domain, followed by a mixture of low vegetation and trees and buildings. Consistent with the phenomena, significant confusion is observed among specific categories, such as the apparent confusion between buildings and impervious surfaces, low vegetation and trees, and impervious surfaces and clutter in Figs. 6 and 7. As we can see, our method can effectively mitigate misclassification among different categories, producing the sharpest edges and best segmentation results even at exceedingly low bit rates.

5) Visualization of Features: As shown in Fig. 2, with a decrease in bit rates, the ambiguity in the feature space increases, leading to significant degradation of semantic segmentation performance. To solve the problem, the fundamental idea of our proposed RSICM is to compensate for the loss of feature discriminability during the compression process. To assess whether our method can reshape the feature space of compressed images, we visualize the learned features of the uncompressed images (left), the images compressed from the GMM model (middle), and the images compressed from our proposed method (right) in Fig. 8. As can be seen, our method decouples the ambiguity among distinct categories in the feature domain, enhances the intraclass compactness and interclass dispersion, and re-establishes the decision boundaries between different categories. This is further supported by the improved segmentation accuracy of the easily confused classes, as depicted in Figs. 6 and 7. This result is consistent with our motivation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:53:04 UTC from IEEE Xplore. Restrictions apply.

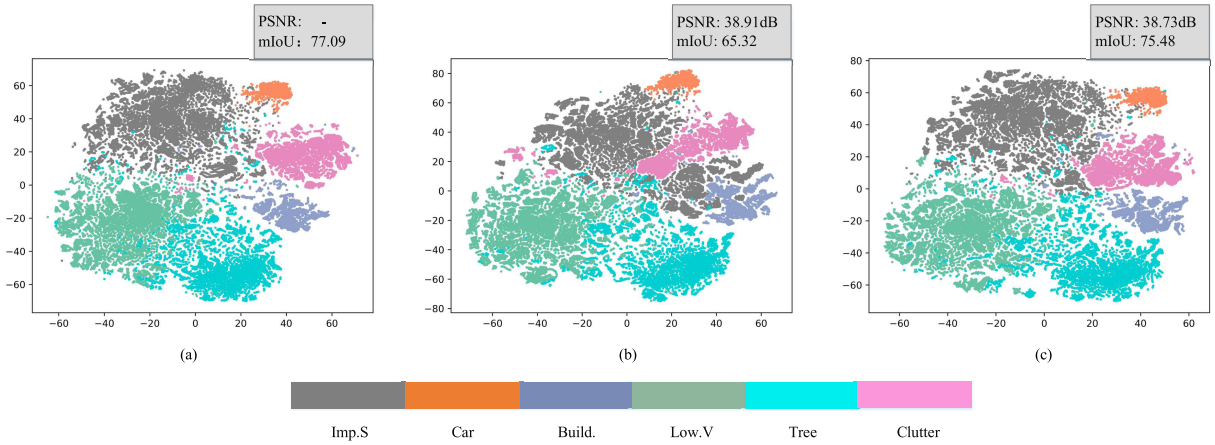


Fig. 8. t-SNE visualization of the learned features. (a) Uncompressed image. (b) Decoded image by the GMM model at 0.45 bpp. (c) Decoded image by our method at 0.45 bpp. PSNR values and mIoU scores are indicated in the upper right corner of the graphics, while distinct colors represent different categories.

TABLE IV
SEGMENTATION PERFORMANCE ON DIFFERENT LAND-COVER CLASSES

Codec	Imp.S	Build.	Low.V.	Tree	Car
UC	86.22	92.65	75.73	78.52	91.08
WebP [10]	43.68	67.01	18.69	18.01	54.26
JPEG2K [11]	39.3	59.43	29.14	36.55	42.31
BPG [13]	33.92	68.27	7.92	32.28	62.92
GDN [19]	33.51	69.01	9.6	27.66	66.8
Hyper [20]	40.18	70.01	12.55	34.1	71.12
Joint [21]	41.51	69.38	13.51	37.43	72.43
JPEGXL [14]	49.55	71.01	34.78	18.84	31.26
GMM [23]	36.21	67.88	25.39	46.7	72.19
Invcomp [27]	34.42	67.39	16.1	46.23	70.83
TBTC [29]	42.0	68.62	36.05	52.76	74.73
SFT [26]	41.49	67.53	36.29	55.15	74.87
ELIC [30]	40.58	67.39	36.42	54.37	73.68
TCM [28]	41.91	68.42	36.47	54.14	74.61
Ours	**72.05**	**86.57**	**47.74**	**56.66**	**84.36**

6) Segmentation Performance on Different Land-Cover Classes: To explore the impact of compression on the segmentation performance for different land-cover classes, IoU values of different land-cover categories under 0.1 bpp are presented in Table IV and visually represented by line charts in Fig. 9. Consistent with prior research, we focused on the five primary foreground classes. As can be seen, the IoU values for impervious surfaces, low vegetation, and trees are relatively low without compression. This may be due to spectrum similarity across these classes, making them easy to confuse [57], [58]. Visual inspection suggests that significant deformation in the tree may further affect its intraclass similarity. After image compression, the segmentation performance of impervious surfaces, low vegetation, and trees significantly declines across nearly all compression algorithms. In addition, traditional codecs such as WebP, JPEG2000, and JPEG-XL significantly impact the segmentation accuracy of the car class more than learning-based algorithms. Combined with the observations in the feature domain as shown in Fig. 2, we speculate that

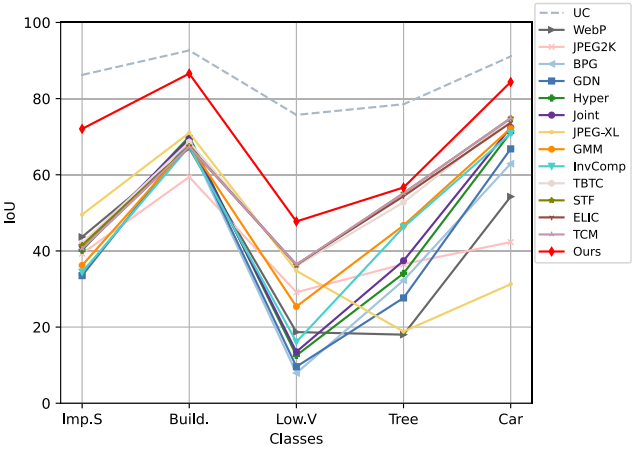


Fig. 9. Comparison results on different land-cover classes.

information loss due to compression may further weaken the already small differences between these categories, leading to the feature ambiguity of different categories, which in turn leads to a sharp degradation in semantic segmentation performance. Based on this, our method addresses the problem from the perspective of enhancing feature discrimination. Specifically, we introduce contrastive learning to guide the compression network to enhance intraclass compactness and interclass dispersion, to preserve the feature discrimination beneficial for semantic segmentation. The experimental results indicate that our method significantly improves segmentation performance for each category compared to the basic GMM model (the orange line in Fig. 9) and outperforms all other comparative compression methods. However, there remains a performance gap in Low Vegetation and Tree classes compared to uncompressed images. This may be due to the inherent difficulty in classifying these two classes, which poses challenges to retaining enough differentiated information to distinguish them well at low bit rates.

V. CONCLUSION

In this article, we proposed a remote sensing ICM framework from a unique perspective of feature discrimination.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:53:04 UTC from IEEE Xplore. Restrictions apply.

Specifically, we innovatively incorporate contrastive learning into the RD optimization process to guide the compression network enhancing intraclass compactness and interclass dispersion, thereby compensating for the feature discrimination distorted by the encoding process. A comprehensive set of benchmark experiments on the ISPRS Potsdam, Vaihingen, and LoveDA datasets demonstrated the effectiveness of the proposed method. Furthermore, the visualization of features confirmed that our method can compensate for the feature discrimination distorted by the compression. However, while our training strategy theoretically has the potential for extension to various visual tasks, it currently necessitates retraining the compression network for each specific task. In future work, we will continue to explore a more general compression framework applicable to multiple downstream vision tasks.

ACKNOWLEDGMENT

The numerical calculations in this article have been done on the supercomputing system in the Supercomputing Center of Wuhan University.

REFERENCES

[1] A. Ma, N. Yu, Z. Zheng, Y. Zhong, and L. Zhang, "A supervised progressive growing generative adversarial network for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5618818.

[2] L. Ding and L. Bruzzone, "DiResNet: Direction-aware residual network for road extraction in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10243–10254, Dec. 2021.

[3] X. Peng, R. Zhong, Z. Li, and Q. Li, "Optical remote sensing image change detection based on attention mechanism and image difference," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7296–7307, Sep. 2021.

[4] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[5] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[6] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.

[7] A. Zabala, C. Cea, and X. Pons, "Segmentation and thematic classification of color orthophotos over non-compressed and JPEG 2000 compressed images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 15, pp. 92–104, Apr. 2012.

[8] A. Zabala, R. Vitulli, and X. Pons, "Impact of CCSDS-IDC and JPEG 2000 compression on image quality and classification," *J. Electr. Comput. Eng.*, vol. 2012, no. 1, Jan. 2012, Art. no. 761067.

[9] G. K. Wallace, "The JPEG still picture compression standard," *Commun. ACM*, vol. 34, no. 4, pp. 30–44, Apr. 1991.

[10] G. Ginesu, M. Pintus, and D. D. Giusto, "Objective assessment of the webp image coding algorithm," *Signal Process., Image Commun.*, vol. 27, no. 8, pp. 867–874, 2012.

[11] M. Rabbani and R. Joshi, "An overview of the JPEG 2000 still image compression standard," *Signal Process., Image Commun.*, vol. 17, no. 1, pp. 3–48, Jan. 2002.

[12] F. Garcia-Vilchez and J. Serra-Sagrista, "Extending the CCSDS recommendation for image data compression for remote sensing scenarios," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 10, pp. 3431–3445, Oct. 2009.

[13] F. Bellard. (2014). *BPG Image Format*. [Online]. Available: http://bellard.org/bpg/

[14] J. Alakuijala et al., "JPEG XL next-generation image compression architecture and coding tools," *Proc. SPIE, Appl. Digit. Image Process. XLII*, vol. 11137, 2019, pp. 112–124, Art. no. 111370K.

[15] B. Penna, T. Tillo, E. Magli, and G. Olmo, "Progressive 3-D coding of hyperspectral images based on JPEG 2000," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 1, pp. 125–129, Jan. 2006.

[16] Q. Du and J. E. Fowler, "Hyperspectral image compression using JPEG2000 and principal component analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 4, no. 2, pp. 201–205, Apr. 2007.

[17] D. Báscones, C. González, and D. Mozos, "Hyperspectral image compression using vector quantization, PCA and JPEG2000," *Remote Sens.*, vol. 10, no. 6, p. 907, Jun. 2018.

[18] W. Zhu, Q. Du, and J. E. Fowler, "Multitemporal hyperspectral image compression," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 3, pp. 416–420, May 2011.

[19] J. Ballé, V. Laparra, and E. P. Simoncelli, "End-to-end optimized image compression," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2017, pp. 1–27.

[20] J. Ballé, D. Minnen, S. Singh, S. Hwang, and N. Johnston, "Variational image compression with a scale hyperprior," in *Proc. 6th Int. Conf. Learn. Rep. (ICLR)*, Apr. 2018, pp. 1–47.

[21] D. Minnen, J. Ballé, and G. D. Toderici, "Joint autoregressive and hierarchical priors for learned image compression," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, Dec. 2018, pp. 10771–10780.

[22] M. Li, W. Zuo, S. Gu, D. Zhao, and D. Zhang, "Learning convolutional networks for content-weighted image compression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3214–3223.

[23] Z. Cheng, H. Sun, M. Takeuchi, and J. Katto, "Learned image compression with discretized Gaussian mixture likelihoods and attention modules," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 7936–7945.

[24] Y. Hu, W. Yang, Z. Ma, and J. Liu, "Learning end-to-end lossy image compression: A benchmark," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 8, pp. 4194–4211, Mar. 2021.

[25] T. Chen, H. Liu, Z. Ma, Q. Shen, X. Cao, and Y. Wang, "End-to-end learnt image compression via non-local attention optimization and improved context modeling," *IEEE Trans. Image Process.*, vol. 30, pp. 3179–3191, 2021.

[26] R. Zou, C. Song, and Z. Zhang, "The devil is in the details: Window-based attention for image compression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 17492–17501.

[27] Y. Xie, K. L. Cheng, and Q. Chen, "Enhanced invertible encoding for learned image compression," in *Proc. 29th ACM Int. Conf. Multimedia*, Oct. 2021, pp. 162–170.

[28] J. Liu, H. Sun, and J. Katto, "Learned image compression with mixed transformer-CNN architectures," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 14388–14397.

[29] Y. Zhu, Y. Yang, and T. Cohen, "Transformer-based transform coding," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021, pp. 1–35.

[30] D. He, Z. Yang, W. Peng, R. Ma, H. Qin, and Y. Wang, "ELIC: Efficient learned image compression with unevenly grouped space-channel contextual adaptive coding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5718–5727.

[31] S. Zhao, S. Yang, J. Gu, Z. Liu, and Z. Feng, "Symmetrical lattice generative adversarial network for remote sensing images compression," *ISPRS J. Photogramm. Remote Sens.*, vol. 176, pp. 169–181, Jun. 2021.

[32] P. Han, B. Zhao, and X. Li, "Edge-guided remote-sensing image compression," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5524515.

[33] S. Xiang and Q. Liang, "Remote sensing image compression with long-range convolution and improved non-local attention model," *Signal Process.*, vol. 209, Aug. 2023, Art. no. 109005.

[34] L. Duan, J. Liu, W. Yang, T. Huang, and W. Gao, "Video coding for machines: A paradigm of collaborative compression and intelligent analytics," *IEEE Trans. Image Process.*, vol. 29, pp. 8680–8695, 2020.

[35] N. Le, H. Zhang, F. Cricri, R. Ghaznavi-Youvalari, and E. Rahtu, "Image coding for machines: An end-to-end learned approach," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2021, pp. 1590–1594.

[36] L. D. Chamain, F. Racape, J. Begaint, A. Pushparaja, and S. Feltman, "End-to-end optimized image compression for machines, a study," in *Proc. Data Compress. Conf.*, Mar. 2021, pp. 163–172.

[37] Y.-H. Chen, Y.-C. Weng, C.-H. Kao, C. Chien, W.-C. Chiu, and W.-H. Peng, "TransTIC: Transferring transformer-based image compression from human perception to machine perception," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 23297–23307.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:53:04 UTC from IEEE Xplore. Restrictions apply.

[38] Z. Chen, K. Fan, S. Wang, L. Duan, W. Lin, and A. C. Kot, "Toward intelligent sensing: Intermediate deep feature compression," *IEEE Trans. Image Process.*, vol. 29, pp. 2230–2243, 2020.

[39] H. Choi and I. V. Bajic, "Deep feature compression for collaborative object detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 3743–3747.

[40] C. Guo and L. Zhang, "A novel multiresolution spatiotemporal saliency detection model and its applications in image and video compression," *IEEE Trans. Image Process.*, vol. 19, no. 1, pp. 185–198, Jan. 2010.

[41] Z. Li, S. Qin, and L. Itti, "Visual attention guided bit allocation in video compression," *Image Vis. Comput.*, vol. 29, no. 1, pp. 1–14, Jan. 2011.

[42] M. Meddeb, M. Cagnazzo, and B. Pesquet-Popescu, "Region-of-interest based rate control scheme for high efficiency video coding," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2014, pp. 7338–7342.

[43] Q. Cai, Z. Chen, D. O. Wu, S. Liu, and X. Li, "A novel video coding strategy in HEVC for object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 12, pp. 4924–4937, Dec. 2021.

[44] W. Wang, T. Zhou, F. Yu, J. Dai, E. Konukoglu, and L. V. Gool, "Exploring cross-image pixel contrast for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7303–7313.

[45] P. K. Verma et al., "Machine-to-machine (M2M) communications: A survey," *J. Netw. Comput. Appl.*, vol. 66, pp. 83–105, May 2016.

[46] Z. Chen, K. Fan, S. Wang, L.-Y. Duan, W. Lin, and A. Kot, "Lossy intermediate deep learning feature compression and evaluation," in *Proc. 27th ACM Int. Conf. Multimedia*, 2019, pp. 2414–2422.

[47] S. Singh, S. Abu-El-Haija, N. Johnston, J. Ballé, A. Shrivastava, and G. Toderici, "End-to-end learning of compressible features," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2020, pp. 3349–3353.

[48] H. Choi and I. V. Bajic, "High efficiency compression for object detection," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2018, pp. 1792–1796.

[49] K. Fischer, F. Brand, C. Herglotz, and A. Kaup, "Video coding for machines with feature-based rate-distortion optimization," in *Proc. IEEE 22nd Int. Workshop Multimedia Signal Process. (MMSP)*, Sep. 2020, pp. 1–6.

[50] K. Fischer, C. Forsch, C. Herglotz, and A. Kaup, "Analysis of neural image compression networks for machine-to-machine communication," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2021, pp. 2079–2083.

[51] K. Fischer, F. Brand, and A. Kaup, "Boosting neural image compression for machines using latent space masking," 2021, *arXiv:2112.08168*.

[52] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent. (MICCAI)*, 2015, pp. 234–241.

[53] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," 2015, *arXiv:1511.07122*.

[54] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "Semantic image segmentation with deep convolutional nets and fully connected CRFs," 2014, *arXiv:1412.7062*.

[55] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[56] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.

[57] L. Mi and Z. Chen, "Superpixel-enhanced deep neural forest for remote sensing image semantic segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 140–152, Jan. 2020.

[58] H. Li, K. Qiu, L. Chen, X. Mei, L. Hong, and C. Tao, "SCAttNet: Semantic segmentation network with spatial and channel attention mechanism for high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 905–909, Mar. 2020.

[59] X. Yang et al., "An attention-fused network for semantic segmentation of very-high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 177, pp. 238–262, Jul. 2021.

[60] F. Xu, M. S. Wong, R. Zhu, J. Heo, and G. Shi, "Semantic segmentation of urban building surface materials using multi-scale contextual attention network," *ISPRS J. Photogramm. Remote Sens.*, vol. 202, pp. 158–168, Aug. 2023.

[61] R. Strudel, R. Garcia, I. Laptev, and C. Schmid, "Segmenter: Transformer for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7262–7272.

[62] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 12077–12090.

[63] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, 2022.

[64] H. Hu, J. Cui, and L. Wang, "Region-aware contrastive learning for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 16291–16301.

[65] G. Bjontegaard, *Calculation of Average PSNR Differences Between RD-Curves*, ITU SG16 document VCEG-M33, 2001.

[66] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," 2021, *arXiv:2110.08733*.

Junxi Zhang is currently pursuing the Ph.D. degree with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China.

Zhenzhong Chen (Senior Member, IEEE) is currently a Professor with Wuhan University, Wuhan, China.

Shan Liu (Fellow, IEEE) is currently a Tencent Distinguished Scientist and the General Manager of the Tencent Media Laboratory, Tencent America, Palo Alto, CA, USA.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:53:04 UTC from IEEE Xplore. Restrictions apply.